

## ARCHITECT-LEVEL OVERVIEW OF THE GEOMETRIC COGNITIVE CONTROLLER

### 0. WHAT YOU ARE BUILDING

A deterministic geometric cognitive controller, powered by:

- sentence-level context
- word-level geometry
- trig-phase dynamics
- combinatorial variable expansion
- routing matrices
- stable controller channels

This is NOT a neural network guessing patterns.

This is a mathematical information engine.

### 1. THE WORD MATRIX (THE CORE MODULE)

Each word  $\rightarrow 4 \times 5 = 20$ -variable matrix.

Per-letter (12 variables):

- alphabet index  $\alpha$
- absolute position  $i$
- normalized position  $p$
- letter score
- centered position  $c$
- prefix ratio  $r$
- suffix ratio  $u$
- derivatives  $d, a$
- symmetry diff  $y$
- bias

Per-word (8 variables):

- score density
- total score
- balance
- roughness
- curvature
- symmetry
- length
- bias

Flattened word core:  $w \in \mathbb{R}^{20}$ .

## 2. WORD EXPANSION NETWORK

Each 20-variable word vector passes through:

- trig expansions
- phase transforms
- normalization
- geometric reshaping

Producing embedding  $z_i = f(w_i)$ .

## 3. SENTENCE MATRIX ( $N \times 20$ )

A sentence of  $N$  words becomes:

$$S \in \mathbb{R}^{N \times 20}$$

This is the sentence geometry.

## 4. SENTENCE-LEVEL VARIABLES (69)

Derived from the entire sentence:

- Aggregates — 41
- Geometry — 8

- Dynamics — 5
- Alphabet/letter spectrum — 6
- Trig-phase — 4
- Sensitivity conditioning — 5

Total: 69 stable contextual variables.

## 5. EXPANSION OF SENTENCE VARIABLES

Each of the 69 variables can be expanded:

Single-variable expansion:  $69 \times 20 = 1380$  variables.

Multi-variable combinations:  $(2^{69} - 1)$  combinations.

Each produces 20 geometry variables.

Total manifold  $\approx 1.18 \times 10^{22}$  variables.

## 6. MULTI-VARIABLE INPUTS

Any combination:

- (v1)
- (v1, v2)
- (v1, v5, v6)
- ... up to all 69

Each produces rich geometry → multi-mode behavior.

## 7. ROUTING VIA MULTIPLICATION MATRIX

After expansion:

$$y = Mx$$

The matrix performs:

- rotation
- compression
- normalization

- gating
- smoothing
- channel selection

Converts raw geometry → stable controller signals.

## 8. DYNAMIC ROUTING (PREVIOUS SENTENCE CONDITIONING)

Routing matrix  $M^k$  is modulated by:

$$M^k = f(c^{k-1})$$

The previous sentence determines how the current one is interpreted.

This forms deterministic contextual attention.

## 9. THE MAIN CONTROLLER

Receives 20–40 stable channels:

- speed
- acceleration
- damping
- thresholds
- gains
- bias
- modes
- trig-phase triggers
- special activators

## 10. COMPLETE SYSTEM ARCHITECTURE

Words → Word Matrix → Expansion → Embeddings

Sentence → 69 Vars → Multi-variable Expansion → Routing Matrix → Controller Inputs

Previous Sentence → Conditioner → Modulation

## 11. WHAT YOU HAVE BUILT

A deterministic geometric cognitive engine:

- geometric
- trigonometric
- symmetric
- multi-scale
- context-driven
- dynamically routed
- expressive
- mathematically stable
- capable of millions of modes
- compressible to 20–40 controller channels